

Physical Chemistry (II)

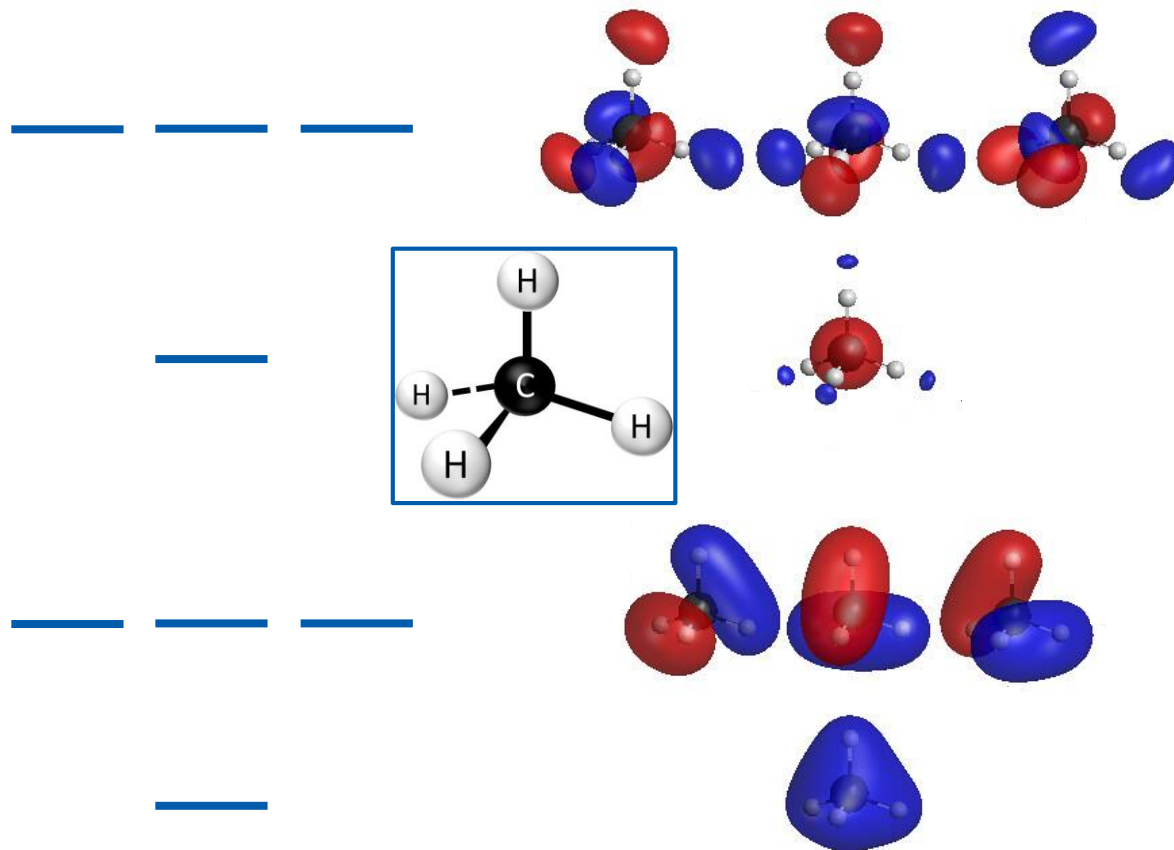
Lecture 23

Polyatomic Molecules (2)



Last class

MOs of Polyatomic Molecules



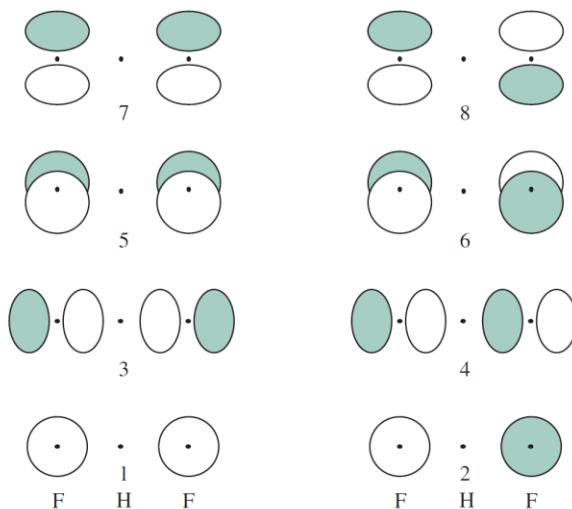
(Image: <https://www.flickr.com/photos/69057297@N04/48054283727>)

Last class

Ingredients to Construct MOs

- Center atom: AOs
- Outer atoms: combinations of AOs = group orbitals

e.g. FHF^- (bifluoride anion)



(Image: Miessler et al., *Inorganic Chemistry*, 5th Ed., Figure 5.15)

Last class

Point Symmetry Operators

a class of symmetry operators
that leave **at least one point unmoved**
(the unmoved point = origin)

Identity operator

Inversion operator

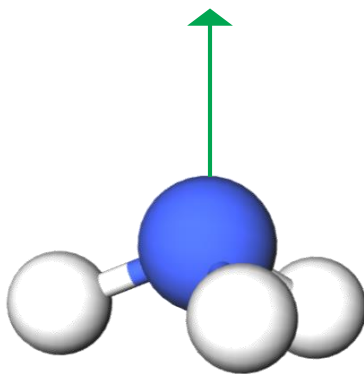
Reflection operator

Rotation operator



Point Group

- A group formed by a set of point symmetry operators belonging to a molecule
- e.g. $\text{NH}_3 - \{\hat{E}, \hat{C}_3, \hat{C}_3^2, \hat{\sigma}_a, \hat{\sigma}_b, \hat{\sigma}_c\}$



Character Table

lists the different symmetry types (“irreps”)
possible in a point group

- Character χ
 - shows how an object or mathematical function is affected by the corresponding symmetry operation of the group
 - 1: no change
 - -1: change of sign
 - otherwise: degenerate case (will be explained soon)

Last class

Overlap Integrals and Irreps

- Overlap integral

$$S_{ij} = \int \psi_i^* \psi_j d\tau$$

For nonzero S_{ij} , ψ_i^* and ψ_j must belong to the same irrep

Projection Operator

$$\hat{P}_j = \sum_{\hat{R}} \chi_j(\hat{R}) \hat{R}$$

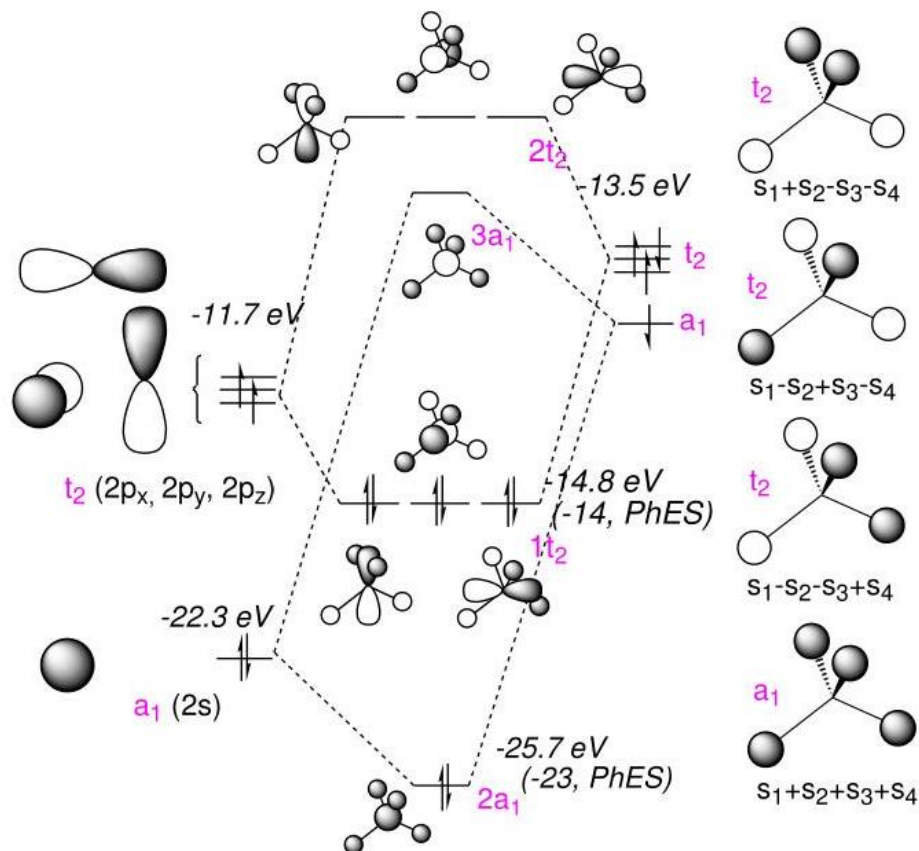
- Useful to find group orbitals belonging to irrep j
- The group orbitals are made of symmetry-adapted linear combinations (SALCs) of atomic orbitals

Today's Topics

1. MO vs. VB theories (example: CH_4): pp. 10-15
2. Potential energy surface (example: H_2O): pp. 16-18
3. Conjugated systems (example: C_2H_4): pp. 19-27
4. Aromaticity and antiaromaticity: pp. 28-30

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MO Diagram of CH₄



(Image: <https://chemistry.stackexchange.com/questions/58774/what-is-responsible-for-making-the-bonds-identical>)

Valence Bond (VB) Theory

H₂ ground state = two ground-state H atoms

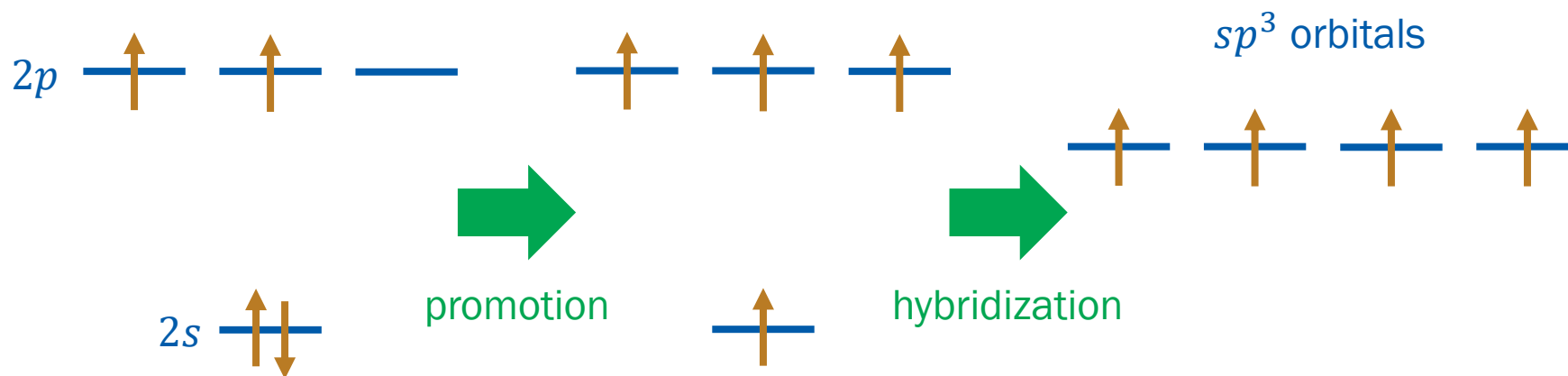
$$\psi_{\text{VB}} = \frac{1}{\sqrt{2(1 + S^2)}} [\psi_{1s,A}(1)\psi_{1s,B}(2) + \psi_{1s,B}(1)\psi_{1s,A}(2)]$$

Can we treat CH₄ by the VB theory?

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VB Theory: Hybridization

- VB Theory requires singly occupied orbitals to form bonds
- Promotion and hybridization



$$\phi_{sp^3} = c_1\psi_{2s} + c_2\psi_{2p_x} + c_3\psi_{2p_y} + c_4\psi_{2p_z}$$

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Hybrid Orbitals

$$\phi_{sp^3} = c_1\psi_{2s} + c_2\psi_{2p_x} + c_3\psi_{2p_y} + c_4\psi_{2p_z}$$

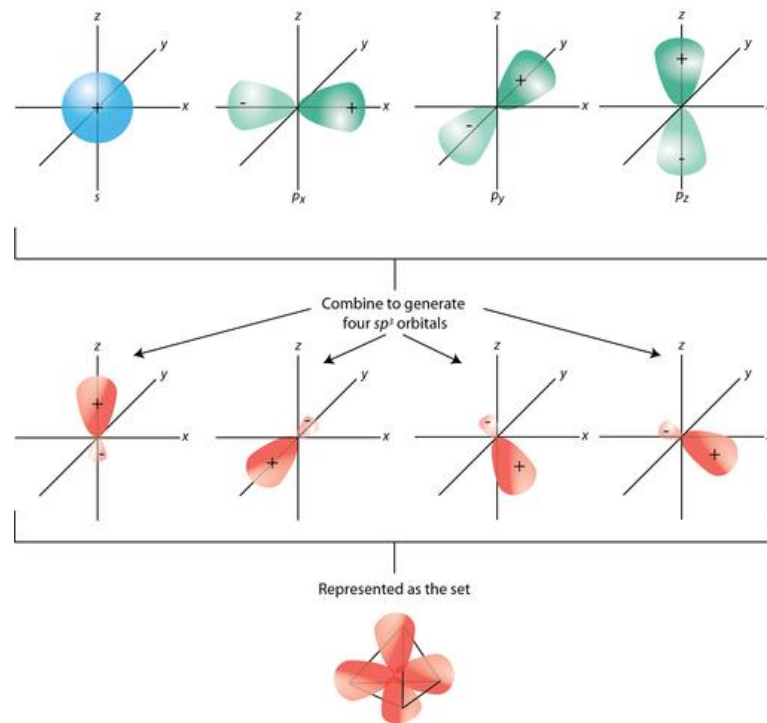
Minimization leads to

$$\phi_{sp^3,1} = \frac{1}{2}(\psi_{2s} + \psi_{2p_x} + \psi_{2p_y} + \psi_{2p_z})$$

$$\phi_{sp^3,2} = \frac{1}{2}(\psi_{2s} - \psi_{2p_x} - \psi_{2p_y} + \psi_{2p_z})$$

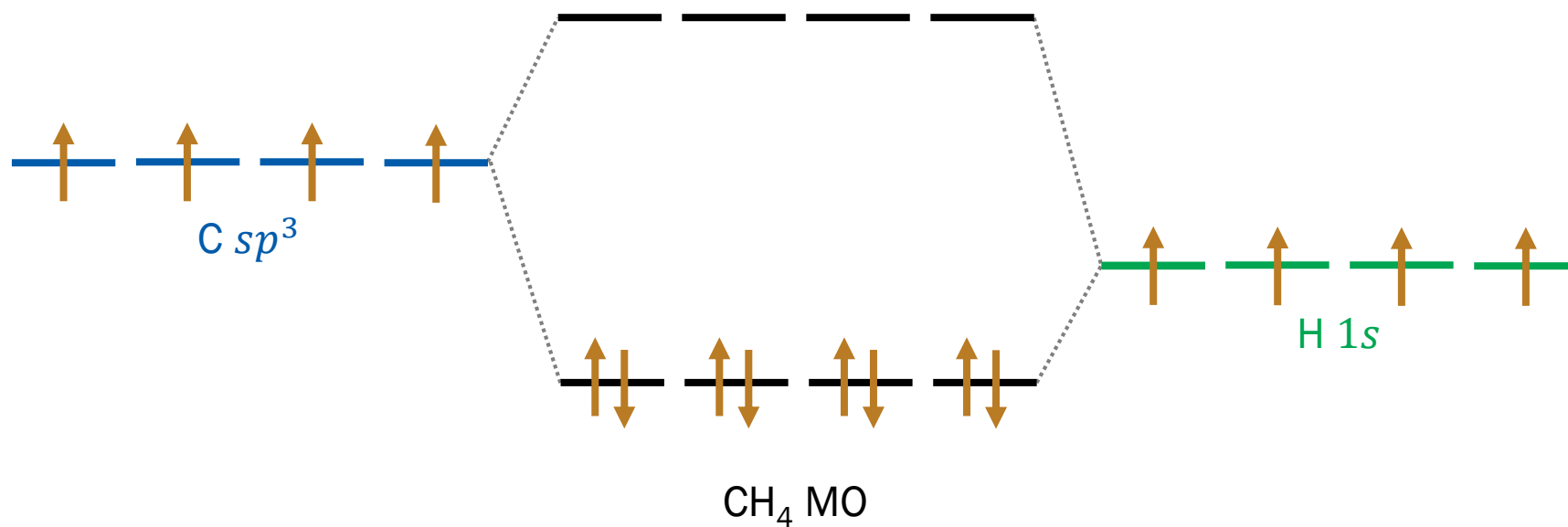
$$\phi_{sp^3,3} = \frac{1}{2}(\psi_{2s} + \psi_{2p_x} - \psi_{2p_y} - \psi_{2p_z})$$

$$\phi_{sp^3,4} = \frac{1}{2}(\psi_{2s} - \psi_{2p_x} + \psi_{2p_y} - \psi_{2p_z})$$



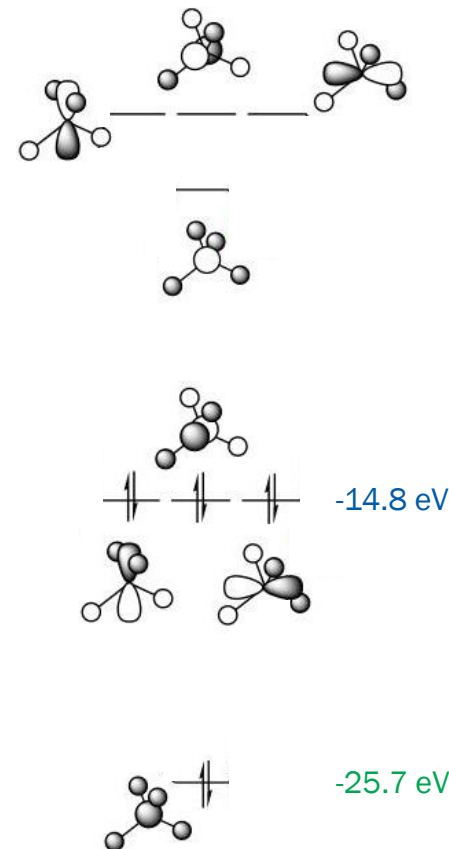
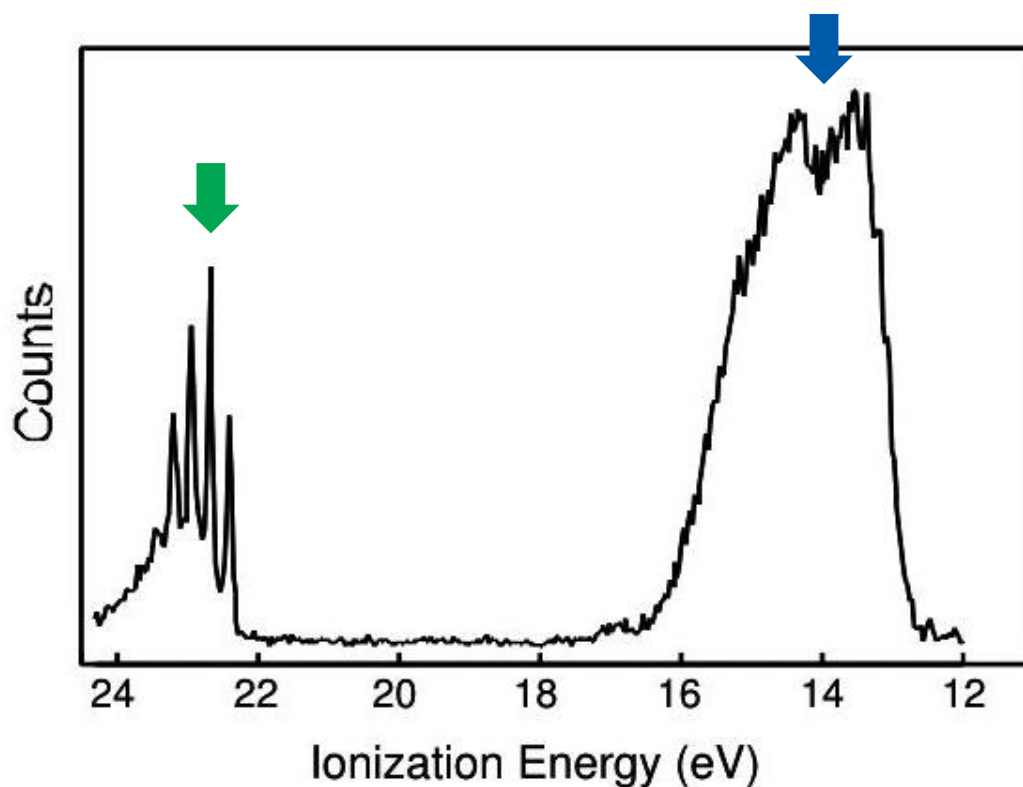
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Construction of MO



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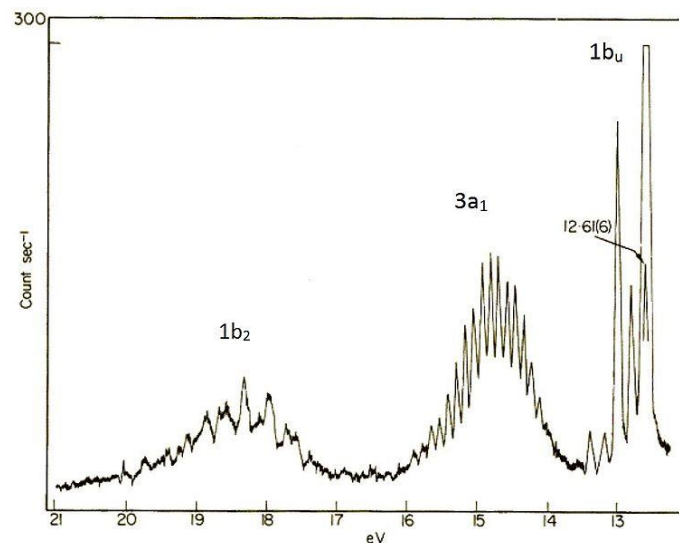
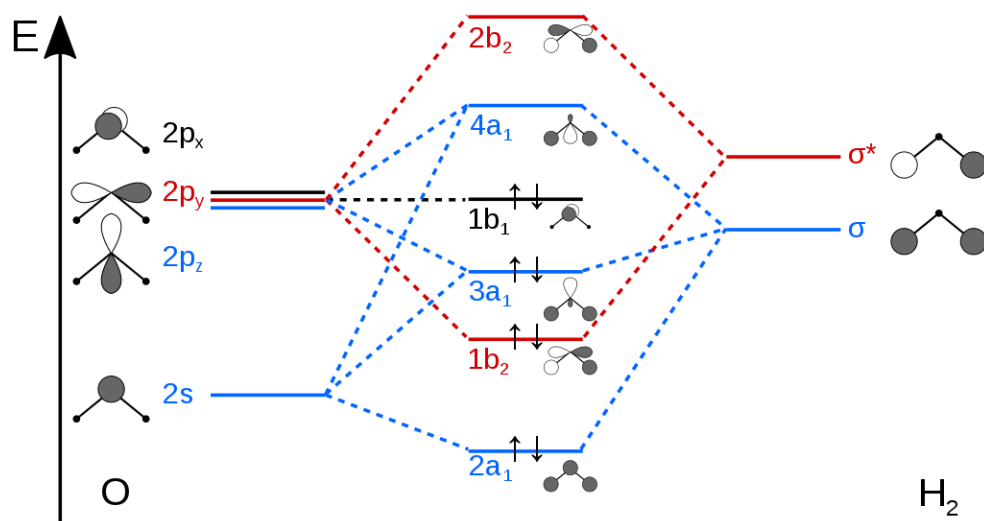
Photoelectron Spectrum of CH₄



(Image: <https://www.slideserve.com/kamal-wiggins/photoelectron-spectroscopy>)

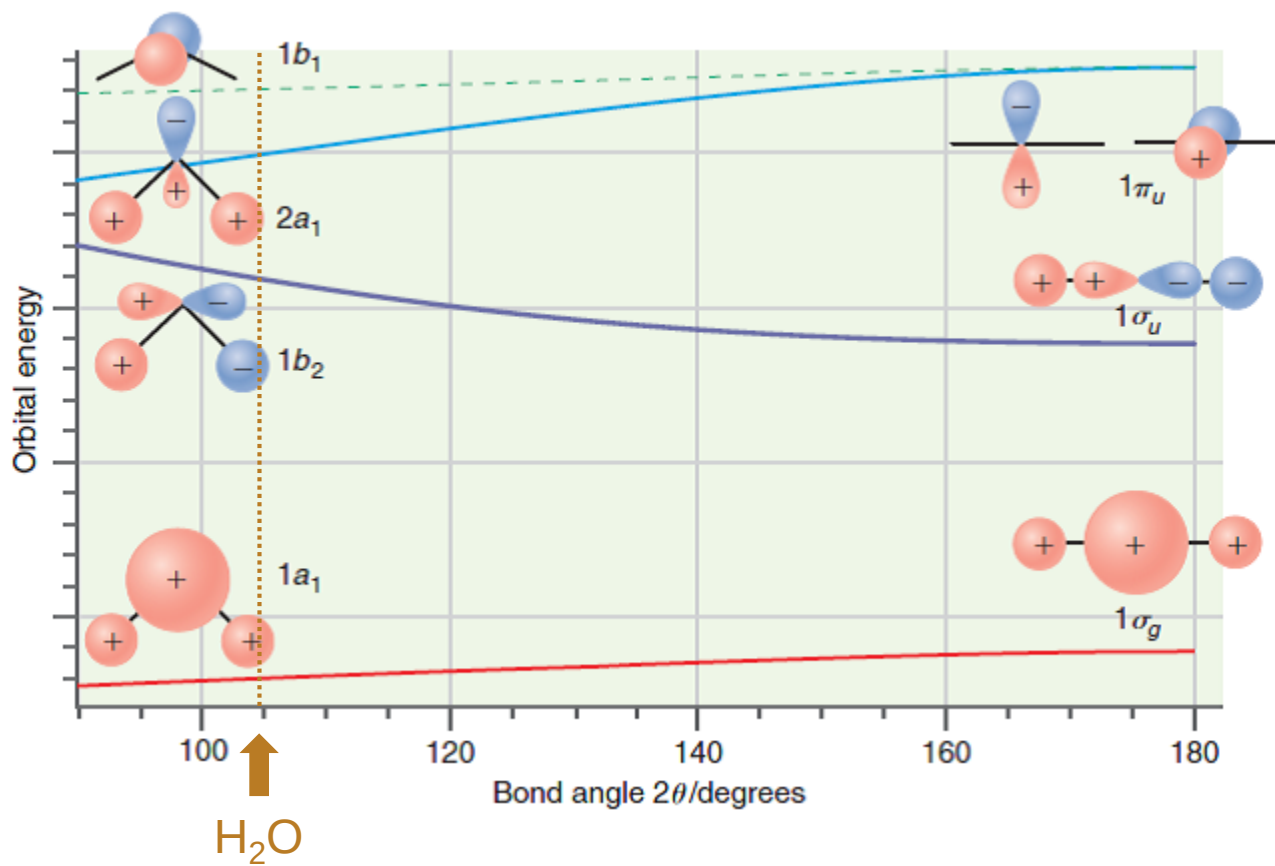
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MO Diagram of H₂O



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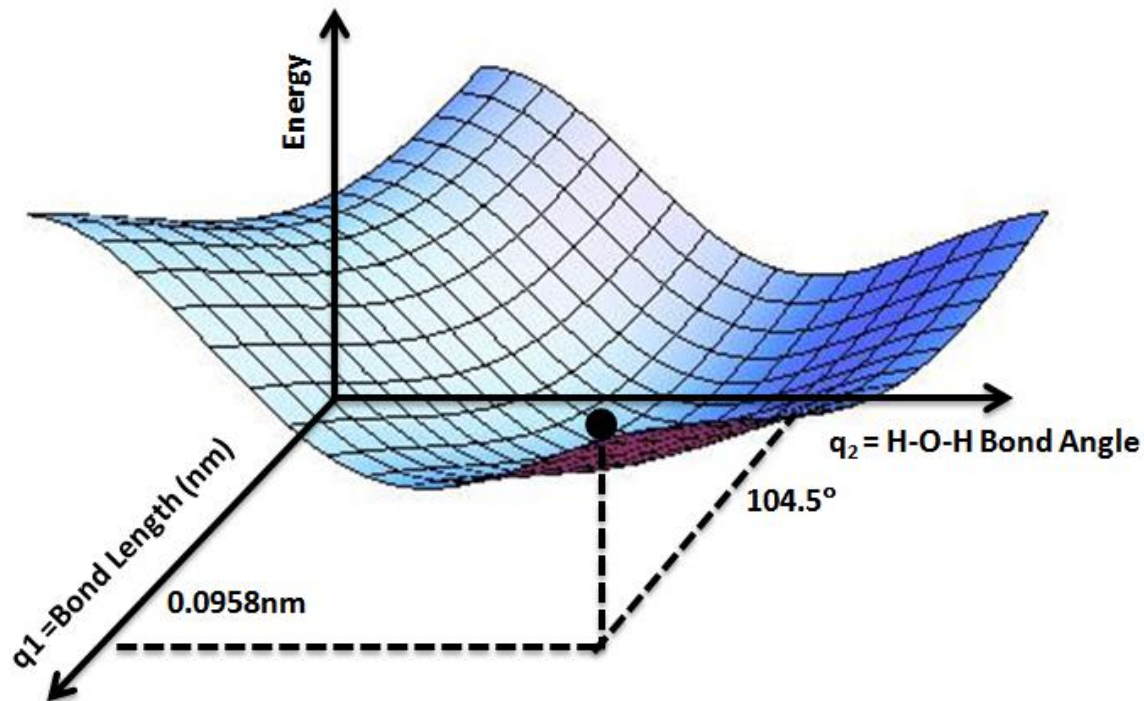
Walsh Diagram: H_2O vs. BeH_2



(Image: <https://www.chegg.com/homework-help/questions-and-answers/using-walsh-diagram-h2o-assume-bond-angle-90o-write-appropriate-expressions-lcaos-involved-q26081156>)

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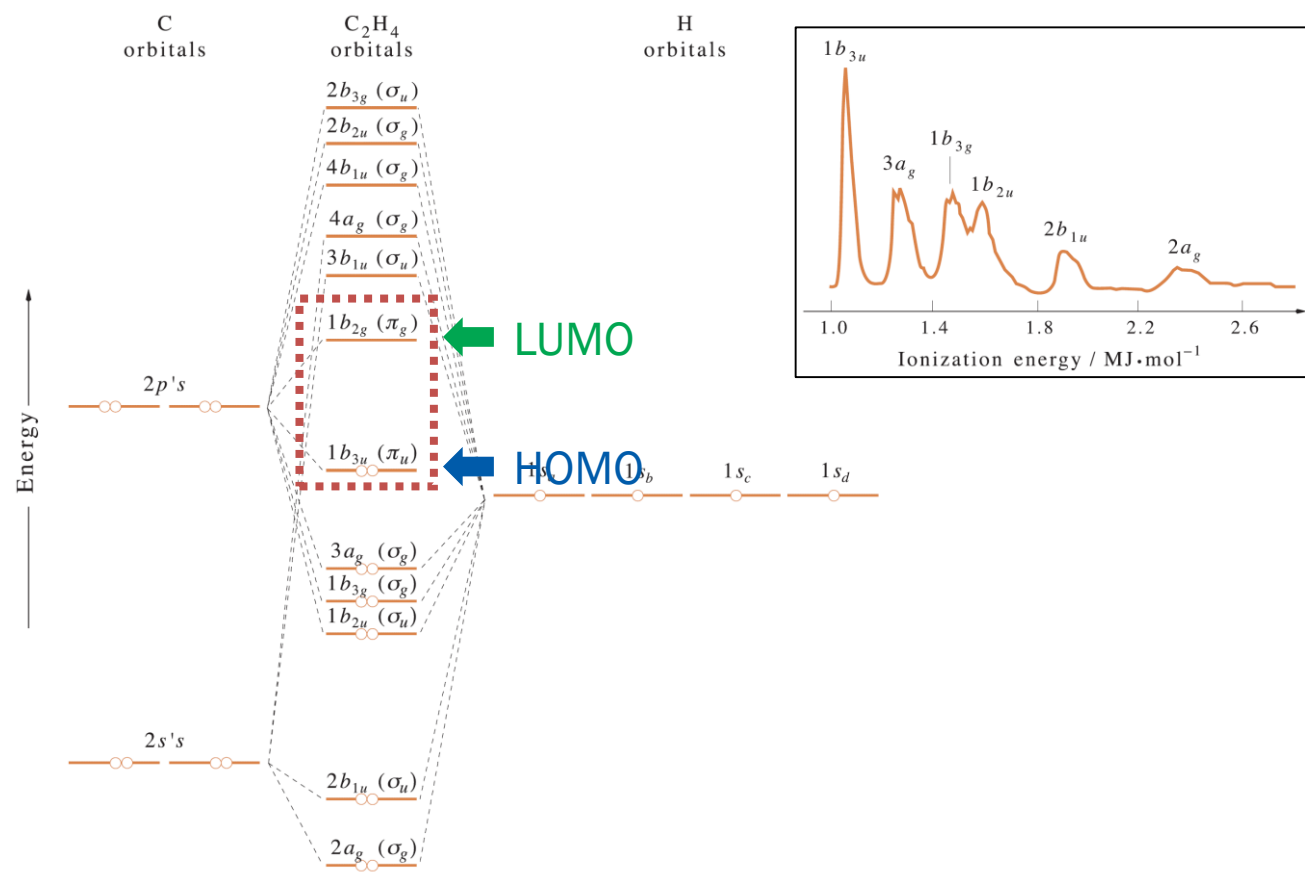
Potential Energy Surface



(Image: https://commons.wikimedia.org/wiki/File:Potential_Energy_Surface_for_Water.png)

RS 12.1
M 11.6

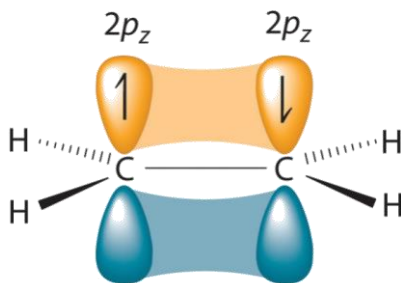
MO Diagram of Ethene (C_2H_4)



(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Figures 11.13 and 11.14)

Conjugated Systems

- **π -electron approximation:** π electrons move in the fixed potential due to the electrons in the σ framework



- Trial function $\phi_{\pi} = c_1\psi_{2p_z,1} + c_2\psi_{2p_z,2}$
- Minimization leads to a secular equation

$$\begin{vmatrix} H_{11} - ES_{11} & H_{12} - ES_{12} \\ H_{21} - ES_{21} & H_{22} - ES_{22} \end{vmatrix} = 0$$

Hückel MO Theory

$$\begin{vmatrix} H_{11} - ES_{11} & H_{12} - ES_{12} \\ H_{21} - ES_{21} & H_{22} - ES_{22} \end{vmatrix} = 0$$

- Simple assumptions

1. Overlap integral $S_{ii} = 1$ and $S_{ij} = 0$
2. Coulomb integral $H_{ii} = \alpha$
3. Exchange integral $H_{ij} = \beta$ for nearest neighbors

$$\begin{vmatrix} \alpha - E & \beta \\ \beta & \alpha - E \end{vmatrix} = 0$$

$$E = \alpha \pm \beta$$

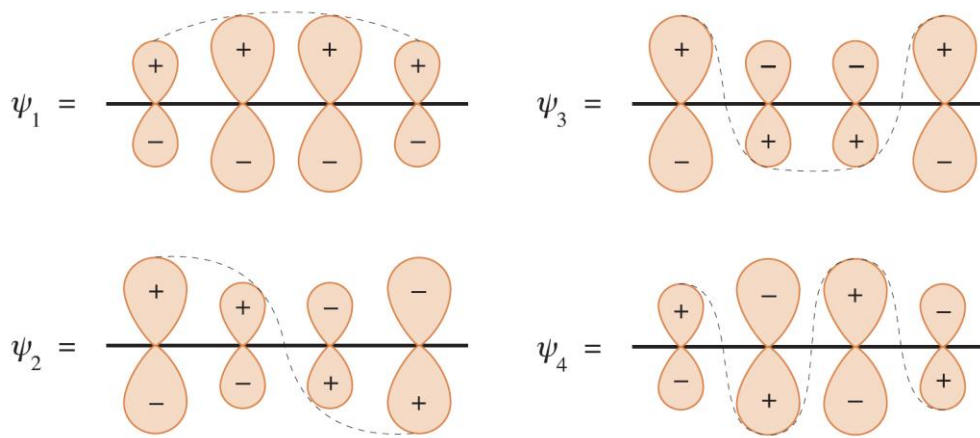
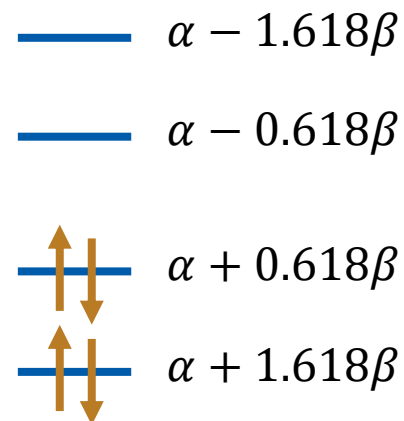
— $\alpha - \beta$

↑↓ $\alpha + \beta$

RS 12.1
M 11.6

Butadiene

$$\begin{vmatrix} \alpha - E & \beta & 0 & 0 \\ \beta & \alpha - E & \beta & 0 \\ 0 & \beta & \alpha - E & \beta \\ 0 & 0 & \beta & \alpha - E \end{vmatrix} = 0$$



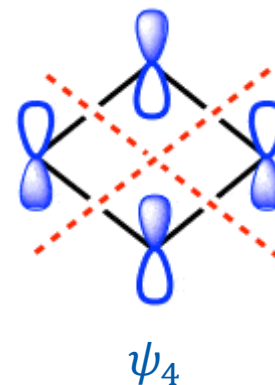
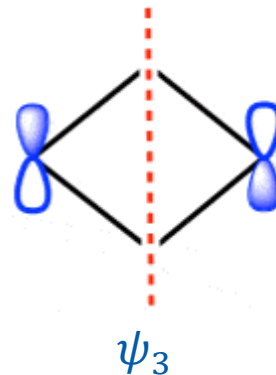
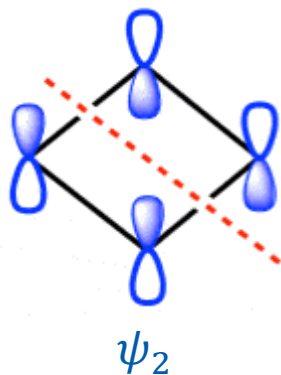
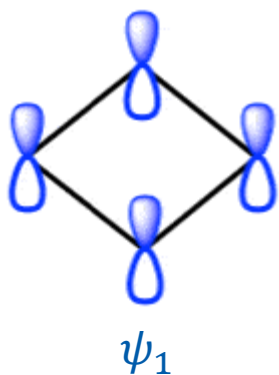
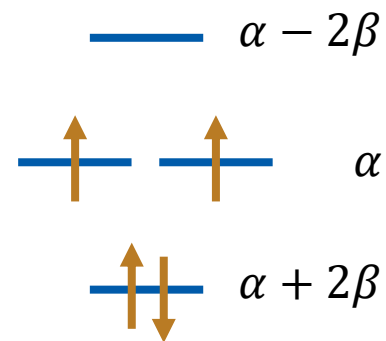
(Image: McQuarrie, *Quantum Chemistry*, 2nd Ed., Figure 11.19)

RS 12.1

M 11.6

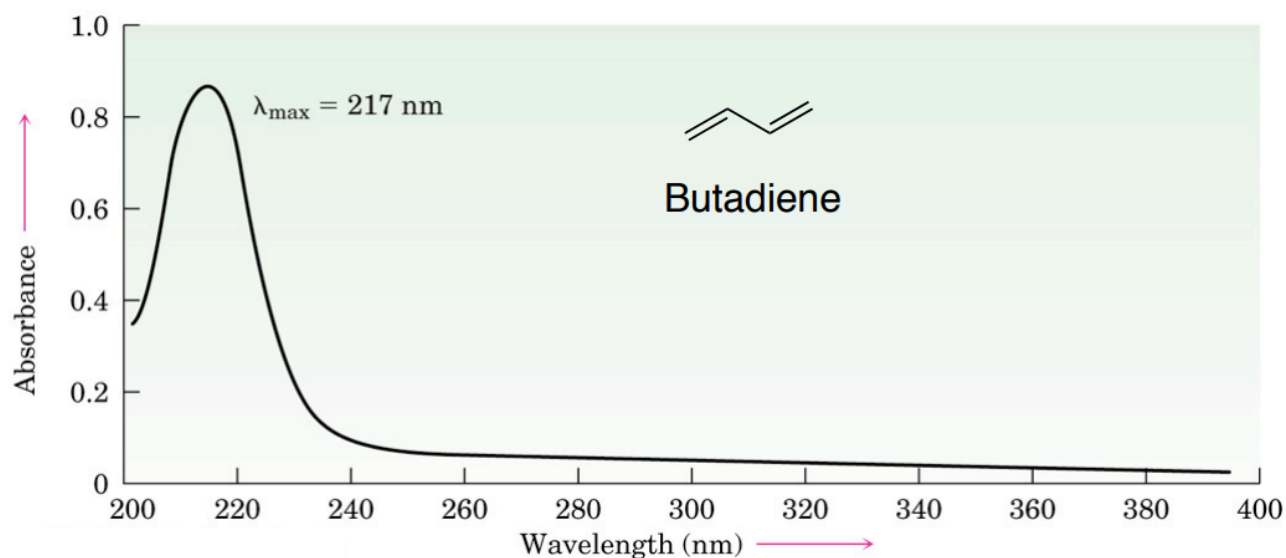
Cyclobutadiene

$$\begin{vmatrix} \alpha - E & \beta & 0 & \beta \\ \beta & \alpha - E & \beta & 0 \\ 0 & \beta & \alpha - E & \beta \\ \beta & 0 & \beta & \alpha - E \end{vmatrix} = 0$$



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UV-Visible Spectroscopy



$$\text{---} \alpha - 1.618\beta$$

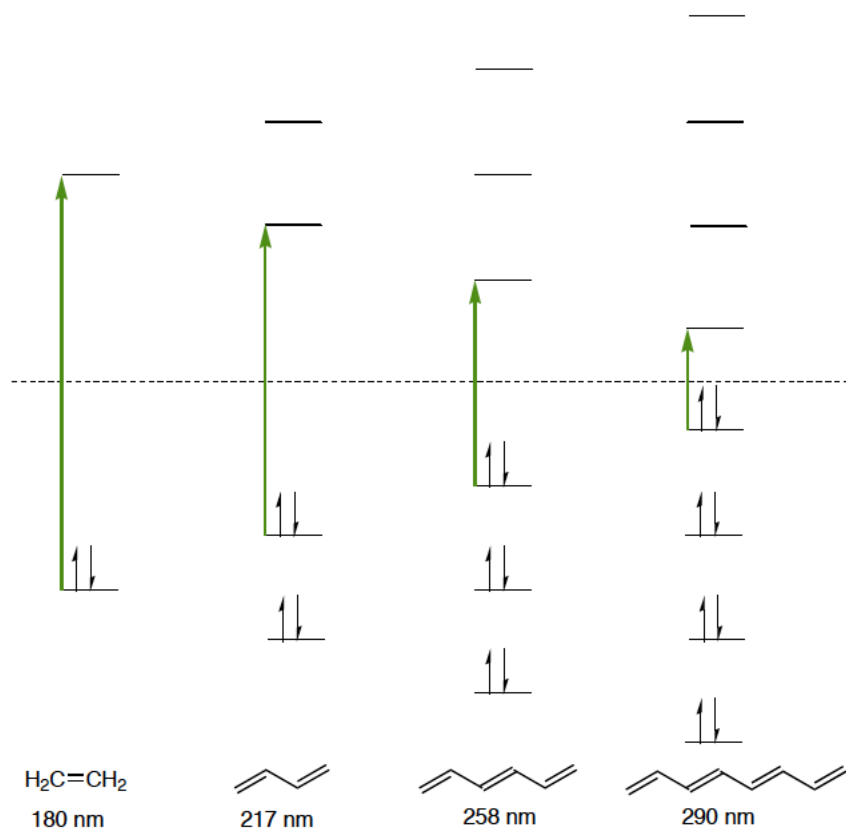
$$\text{---} \alpha - 0.618\beta$$

$$\begin{array}{c} \uparrow \downarrow \\ \text{---} \end{array} \alpha + 0.618\beta$$

$$\begin{array}{c} \uparrow \downarrow \\ \text{---} \end{array} \alpha + 1.618\beta$$

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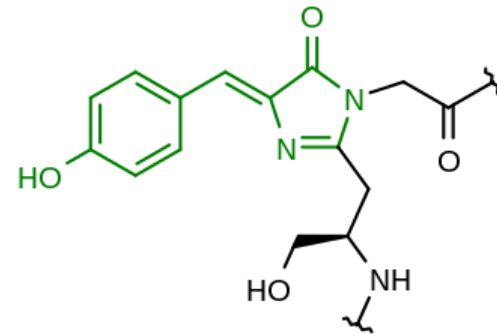
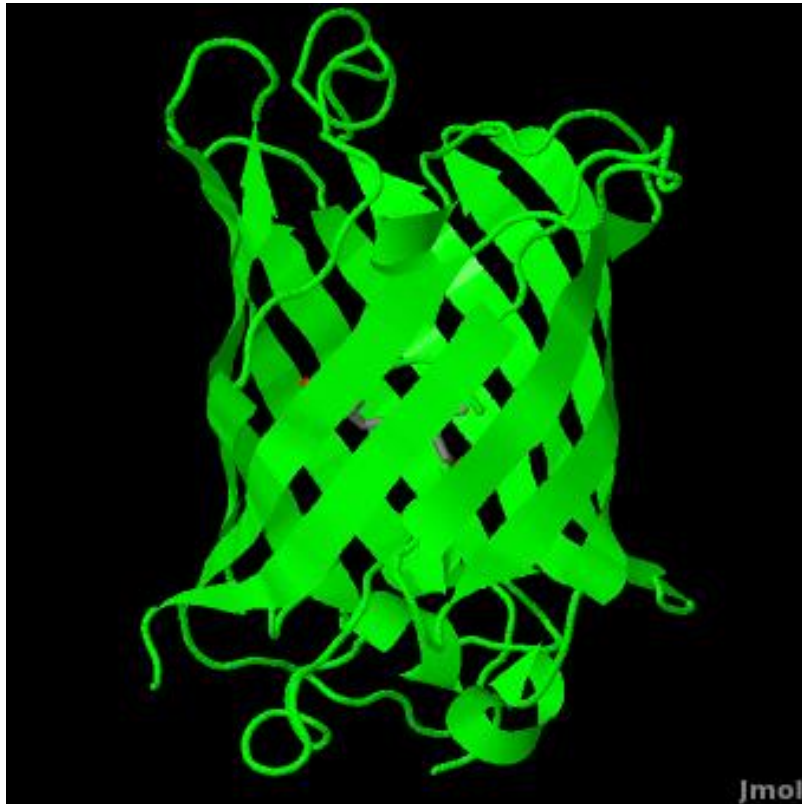
Size of Conjugate System



(Image: <https://www.vanderbilt.edu/AnS/Chemistry/Rizzo/chem220a/Ch14slides.pdf>)

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Green Fluorescent Protein

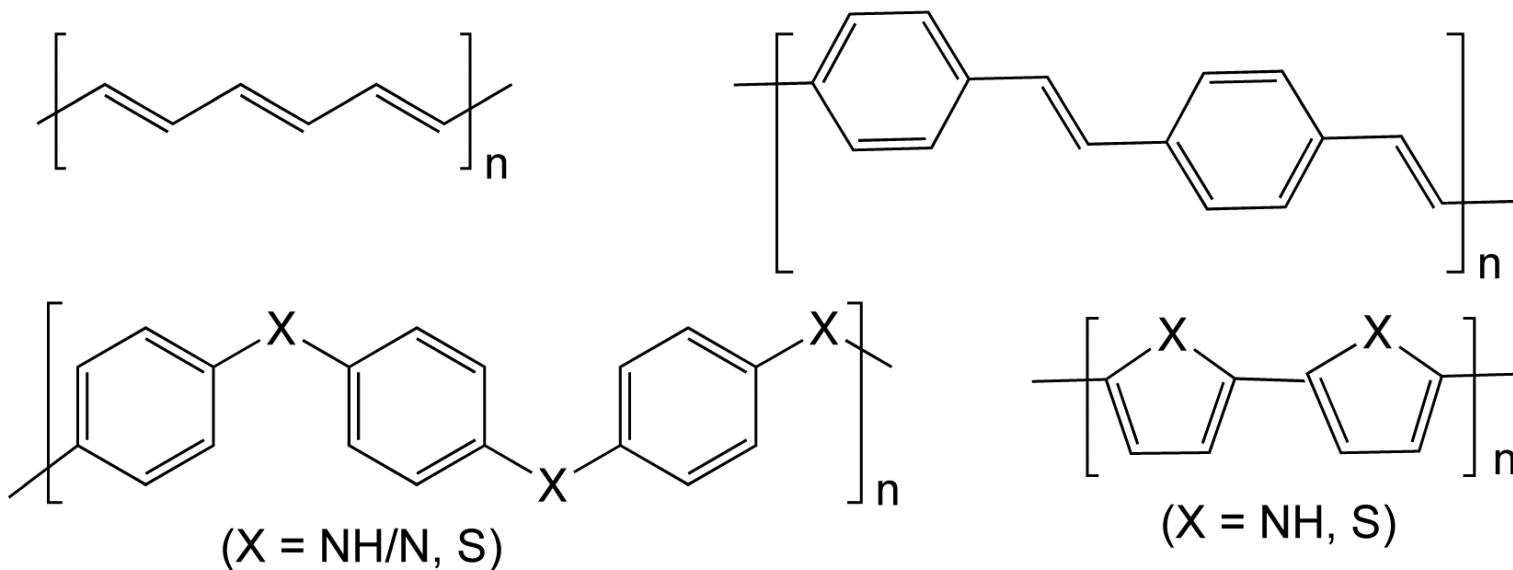


Nobel prize in Chemistry 2008

(Image: https://proteopedia.org/wiki/index.php/Green_Fluorescent_Protein
https://commons.wikimedia.org/wiki/File:GFP_mechanism.svg)

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Conductive Polymers



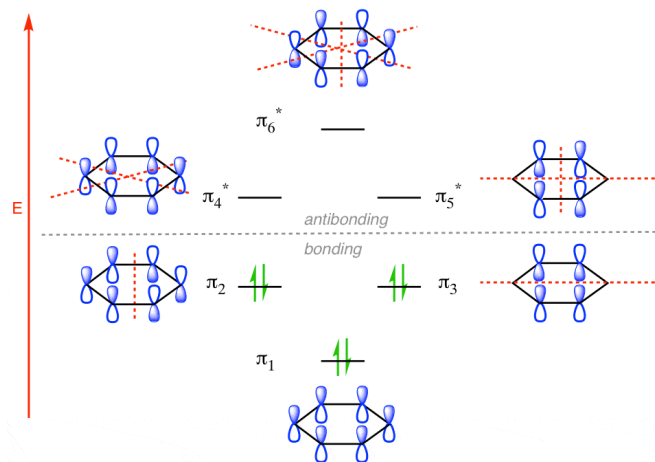
Nobel prize in Chemistry 2000

RS 12.1

M 11.7

Benzene

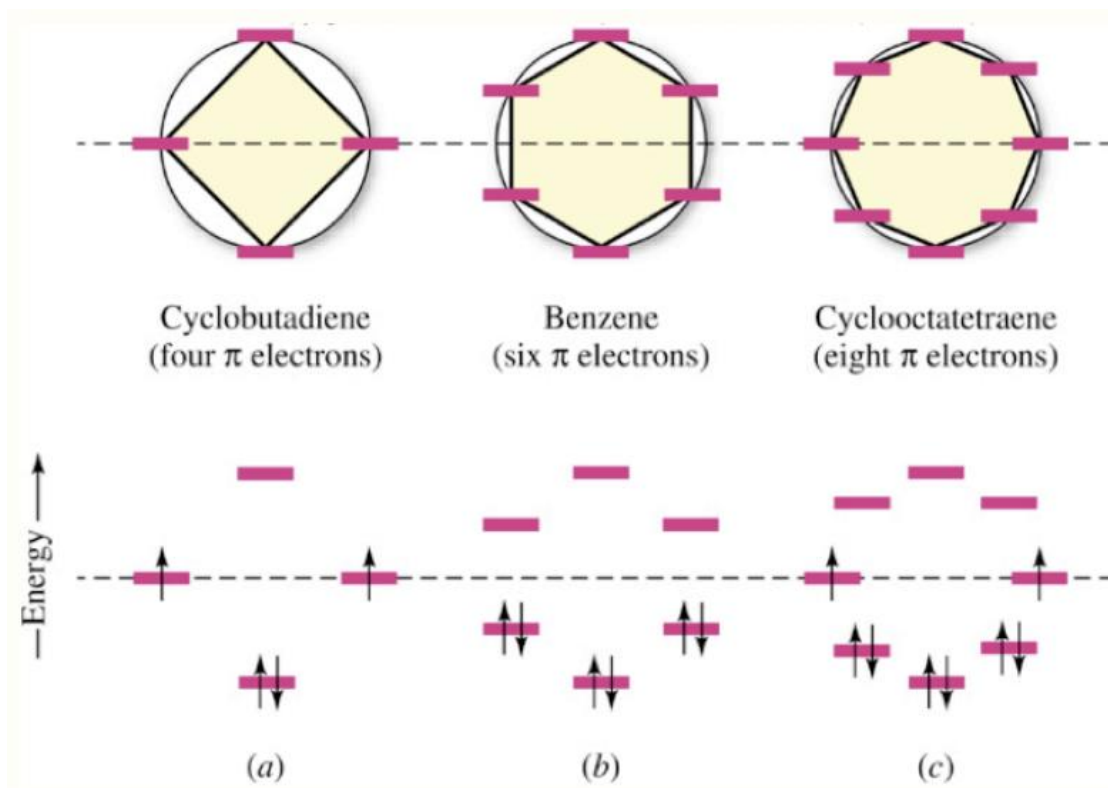
$$\begin{vmatrix} \alpha - E & \beta & 0 & 0 & 0 & \beta \\ \beta & \alpha - E & \beta & 0 & 0 & 0 \\ 0 & \beta & \alpha - E & \beta & 0 & 0 \\ 0 & 0 & \beta & \alpha - E & \beta & 0 \\ 0 & 0 & 0 & \beta & \alpha - E & \beta \\ \beta & 0 & 0 & 0 & \beta & \alpha - E \end{vmatrix} = 0$$



(Image: <https://www.masterorganicchemistry.com/2017/05/05/the-pi-molecular-orbitals-of-benzene/>)

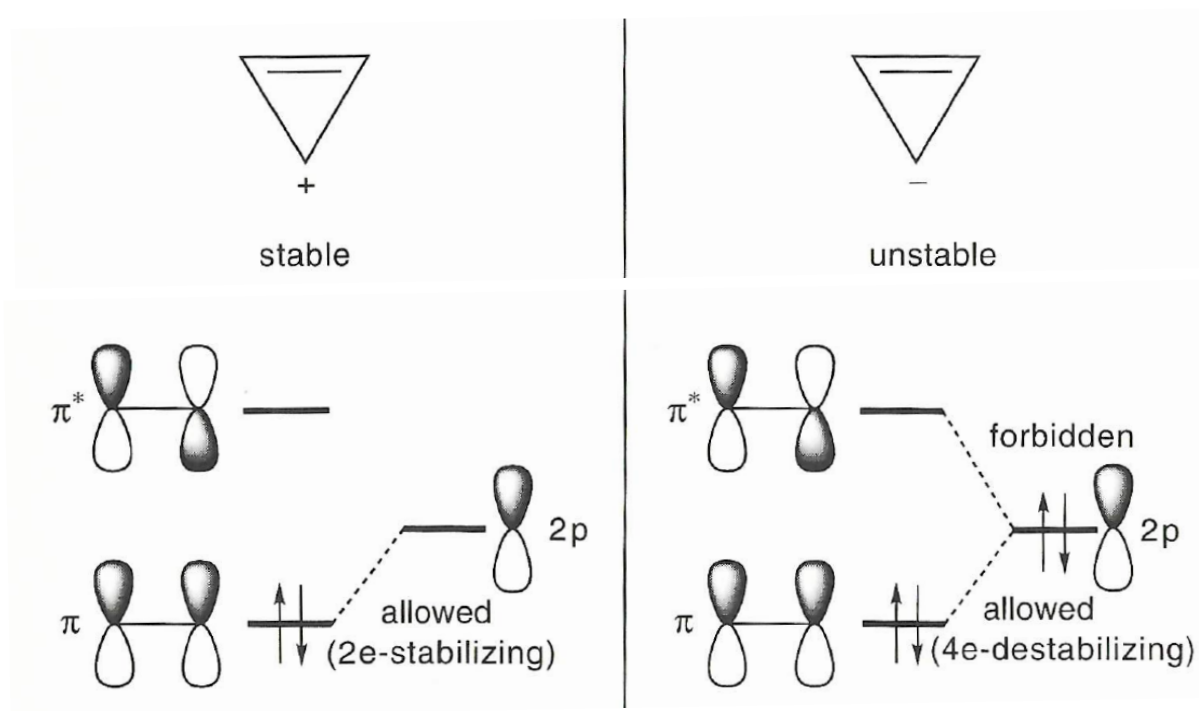
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Aromaticity & Antiaromaticity



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Symmetry Argument



(Image: Pross, *Theoretical & Physical Principles of Organic Reactivity*, Figure 2.26)